

Fig. 3: Actual-size PCB pattern of the low-cost air-quality meter

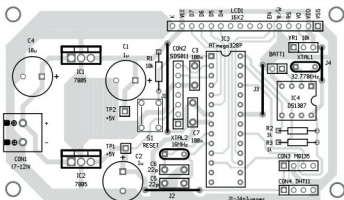


Fig. 4: Component layout of the PCB

tion. UART port of SDS011 (CON2) is connected with IC3 UART port.

12V power/battery supply (CON1) is regulated to 5V using IC1 and IC2 and applied to the circuit. When the system is powered on, the device takes one minute to preheat the gas sensor before it is ready for operation. LCD1 displays proximate values of the four sensors connected to IC3. All these parameters are displayed on LCD1 at an interval of one second.

Software

The software is written in Arduino programming language. ATmega328P is programmed using Arduino IDE software. ATmega328P comes with a bootloader that allows to upload a new code to it without the use of an external hardware programmer. Select the correct board from Tools→Board menu in Arduino

IDE and burn the program (sketch) through the standard USB port in the computer.

Various operations are implemented in the code as follows:

RTC library is easy to use and can get the date, time and day of the week accurately.

DHT.temperature and DHT.humidity read temperature and humidity, respectively.

PM_CAL() function calibrates PM2.5/PM10 value.

The arrangement given below gives a rough approximation of the parts per million (ppm) levels of

TABLE II
TEST POINTS

Test point	Details
TP1, TP2	5V

(Voltage measured with respect to ground)

gas or pollutants. These values are calculated as:

```
count=analogRead(GasSensorPin);
Vout=(count*4.88)/1000.0;
Rs=((5.0*R1)-(R1*Vout))/Vout;
ratio=Rs/approx;
GasConc=146.15*(2.868-ratio)+10;
ppm=GasConc;
```

Construction and testing

An actual-size, single-side PCB for the air-quality meter is shown in Fig. 3 and its component layout in Fig. 4. Suitable connectors are provided on the PCB for input and output.

Assemble the circuit on the PCB to minimise errors. Carefully assemble the components and double-check for overlooked errors. Connect the power supply through 12V DC jack/battery and provide power supply to the circuit by connecting its Vcc and GND to CON1 on the PCB.

To test the circuit for proper functioning, check the power supply at TP1 and TP2 with respect to GND and ensure it is 5V.

All the mentioned parameters (PM2.5/PM10, temperature/humidity, ppm and time/date) are displayed on LCD1 at an interval of one second. In case the text is not visible on LCD1, adjust the potentiometer (VR1) connected with LCD1.

You may use cigarette smoke as polluted air for testing. PM value can reach up to 999 within five seconds when smoke is generated. The ppm value increases with smoke density, and temperature and humidity change as per the environment. **EFY**

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



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